

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems e

Assessment Tool Weightage: 35%

QUESTIONS: 5

TOTAL MARKS;50

Annexure

Analytical Problem Solving

Code of Conduct:

I ARUNPRASATH R (192524077) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARUNPRASATH R

Analytical Problem Solving

1. Bridge and Torch Problem (State-space formulation and UCS)

Four people must cross a narrow bridge at night. They have one torch, and the bridge holds at most two people at a time. Anyone crossing must carry the torch, and a pair walks at the slower person's pace. Their crossing times are 1, 2, 5 and 10 minutes.

- i. Formulate the problem: state representation, initial state, actions, transition model, goal test and path cost.
- ii. Use Uniform Cost Search to find a schedule that gets everyone across in the minimum total time.
- iii. Explain why the obvious strategy (the fastest person escorts everyone) is not optimal.

ANS:

Four people have crossing times:

- Person A = 1 minute
- Person B = 2 minutes
- Person C = 5 minutes
- Person D = 10 minutes

Only two people can cross at a time, and the torch must always be carried.

i. Problem Formulation:

State Representation:

Represent a state as:

(Left side people, Right side people, Torch position)

For example:

({1,2,5,10}, {}, L)

means all four people are on the left and the torch is on the left.

Initial State:

({1,2,5,10}, {}, L)

Goal State:

({}, {1,2,5,10}, R)

All people must be on the right side with the torch.

Actions:

If the torch is on the left:

- One or two people cross from left to right.
- Their crossing time is the slower person's time.

If the torch is on the right:

- One or two people return from right to left with the torch.
- Their cost is the slower person's time.

Transition Model:

For example:

(1,2,5,10 | $\emptyset$, L)

1 and 2 cross:

(5,10 | 1,2, R)

Cost = **2 minutes**.

Goal Test:

Check whether:

Left side = $\emptyset$

and all four people are on the right.

Path Cost:

The total cost is the sum of the crossing times of all actions.

ii. Uniform Cost Search:

UCS always expands the state having the **lowest accumulated cost $g(n)$** .

The optimal sequence is:

Step	Action	Cost	Total
1	1 and 2 cross	2	2
2	1 returns	1	3
3	5 and 10 cross	10	13
4	2 returns	2	15
5	1 and 2 cross	2	17

Therefore:

Optimal schedule:

1,2 $\rightarrow$ 2 min

1 $\leftarrow$ 1 min

5,10 $\rightarrow$ 10 min

2 $\leftarrow$ 2 min

1,2 $\rightarrow$ 2 min

Minimum total time:

$$2 + 1 + 10 + 2 + 2 = \boxed{17 \text{ minutes}}$$

iii. Why is the obvious strategy not optimal?

The obvious strategy is to make the fastest person, 1-minute person, escort everyone.

For example:

- 1 & 10 cross = 10
- 1 returns = 1
- 1 & 5 cross = 5
- 1 returns = 1
- 1 & 2 cross = 2

Total:

$$10 + 1 + 5 + 1 + 2 = \boxed{19}$$

So this strategy takes 19 minutes, whereas UCS finds a solution of 17 minutes.

The key idea is that the 1-minute and 2-minute people should work together to transport the torch efficiently, while the 5- and 10-minute people cross together.

2. Hospital Delivery Robot (PEAS and environment)

A hospital uses an autonomous robot to deliver medicines and lab samples between wards, the pharmacy and the laboratory. It travels through corridors and uses lifts.

- i. Give the PEAS description (Performance measure, Environment, Actuators, Sensors).
- ii. Classify the environment: fully or partially observable, deterministic or stochastic, episodic or sequential, static or dynamic, discrete or continuous, single-agent or multi-agent. Justify each choice.
- iii. Which agent architecture suits this robot? Justify your answer.

ANS:

i. PEAS Description:

PEAS	Description
Performance Measure	Fast and safe delivery, correct destination, minimum travel time, avoiding collisions, efficient battery usage
Environment	Hospital corridors, wards, pharmacy, laboratory, lifts, patients, doctors, nurses and other moving objects
Actuators	Wheels/motors, steering, brakes, lift controls, speaker/display, robotic delivery compartment
Sensors	Cameras, LiDAR/ultrasonic sensors, obstacle sensors, location sensors, RFID/GPS/indoor positioning, lift-status sensors

PEAS:

Performance: Safe, accurate and timely delivery.

Environment: Hospital building containing corridors, rooms, lifts and people.

Actuators: Motors, wheels, brakes, lift interface and communication devices.

Sensors: Camera, LiDAR, obstacle sensors and location sensors.

ii.Environment Classification:

1. Fully or Partially Observable?

Partially Observable

The robot cannot observe everything at all times. People, objects and events may be hidden around corners or inside rooms.

2. Deterministic or Stochastic?

Stochastic

The environment can change unpredictably. For example, a person may suddenly walk into the robot's path or a lift may become unavailable.

3. Episodic or Sequential?

Sequential

Each decision affects future decisions. For example, choosing a particular corridor affects the remaining route and delivery time.

4. Static or Dynamic?

Dynamic

People, carts and other robots may move while the robot is operating.

5. Discrete or Continuous?

Continuous

The robot moves through physical space and continuously senses position, distance and obstacles.

6. Single-Agent or Multi-Agent?

Multi-Agent

Other people, hospital staff, patients and potentially other robots interact with or affect the robot's environment.

Summary

Property	Classification
Observability	Partially observable
Determinism	Stochastic
Episodic/Sequential	Sequential
Static/Dynamic	Dynamic
Discrete/Continuous	Continuous
Agents	Multi-agent

iii. Suitable Agent Architecture

The best choice is a **Model-Based Utility-Based Agent**, possibly combined with **Learning**.

Justification

The robot needs to:

- Maintain an internal model of the hospital.
- Handle partially observable situations.
- Select the safest and fastest route.
- Consider multiple factors such as distance, obstacles, lift availability and battery.
- Adapt to changing conditions.

Therefore, a model-based utility-based architecture is suitable.

4.Route Finding with A* Search

A navigation app has to find the shortest route from S to G. The road costs are:

Edge	Cost	Edge	Cost
S–A	2	B–D	3
S–B	5	C–D	1
A–C	4	C–G	6
Edge	Cost	Edge	Cost
A–D	7	D–G	2

The heuristic values are $h(S)=7$, $h(A)=6$, $h(B)=4$, $h(C)=3$, $h(D)=2$ and $h(G)=0$. i. Apply A* search and show the open and closed lists at each step.
 ii. Give the path found and its cost. iii. Check whether the heuristic is admissible and consistent. iv. Apply Greedy Best-First Search to the same graph and compare the results.

ANS:

i. A* Search — Open and Closed Lists:

Step 0

Start at S.

$$g(S) = 0, f(S) = 0 + 7 = 7$$

OPEN: S(7)

CLOSED: —

Step 1 — Expand S

From S:

- A: $g=2$, $f=2+6=8$
- B: $g=5$, $f=5+4=9$

OPEN: A(8), B(9)

CLOSED: S

Step 2 — Expand A

From A:

- C: $g=2+4=6$, $f=6+3=9$
- D: $g=2+7=9$, $f=9+2=11$

OPEN: B(9), C(9), D(11)

CLOSED: S, A

Step 3 — Expand B

From B:

D can be reached with: $g(D) = 5 + 3 = 8$

This is better than the previous $g(D)=9$.

Therefore:

$$f(D) = 8 + 2 = 10$$

OPEN: C(9), D(10)

CLOSED: S, A, B

Step 4 — Expand C

From C:

D can be reached with: $g(D) = 6 + 1 = 7$

This is better than 8.

Therefore: $f(D) = 7 + 2 = 9$

G can be reached with:

$$\begin{aligned}g(G) &= 6 + 6 = 12 \\f(G) &= 12\end{aligned}$$

OPEN: D(9), G(12)

CLOSED: S, A, B, C

Step 5 — Expand D

From D: $g(G) = 7 + 2 = 9$

Thus: $f(G) = 9 + 0 = 9$

OPEN: G(9)

CLOSED: S, A, B, C, D

Step 6 — Expand G

G is the goal.

CLOSED: S, A, B, C, D, G

ii. Path and Cost:

Tracing the lowest-cost parents: $S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$

Cost:

$$\begin{aligned}S - A &= 2 \\A - C &= 4 \\C - D &= 1 \\D - G &= 2\end{aligned}$$

Therefore: $2 + 4 + 1 + 2 = \boxed{9}$

Final Answer

$$\boxed{S \rightarrow A \rightarrow C \rightarrow D \rightarrow G}$$

Total cost = 9

iii. Is the heuristic admissible and consistent?

Admissibility

A heuristic is admissible if:

$$h(n) \leq h^*(n)$$

where $h^*(n)$ is the actual minimum cost to the goal.

Actual minimum costs:

Node	Actual cost to G	h(n)
S	9	7
A	7	6
B	5	4
C	3	3
D	2	2
G	0	0

Every heuristic value is less than or equal to the actual cost.

Therefore, **the heuristic is admissible**.

Consistency

A heuristic is consistent if:

$$h(n) \leq c(n, n') + h(n')$$

For example:

$$S \rightarrow A: 7 \leq 2 + 6 = 8$$

C $\rightarrow$ D:

$$3 \leq 1 + 2 = 3$$

D $\rightarrow$ G:

$$2 \leq 2 + 0 = 2$$

All edges satisfy the condition.

Therefore:

The heuristic is consistent

iv. Greedy Best-First Search:

Greedy Best-First Search uses: $f(n) = h(n)$

It ignores the path cost $g(n)$.

Step 1

From S:

- A: $h=6$
- B: $h=4$

Choose **B** because $h(B)=4$ is smaller.

Step 2

From B:

- D: $h=2$

Choose **D**.

Step 3

From D:

- G: $h=0$

Choose **G**.

Therefore: $S \rightarrow B \rightarrow D \rightarrow G$

$$\text{Cost: } 5 + 3 + 2 = \boxed{10}$$

Comparison

Algorithm	Path	Cost
A*	$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	9
Greedy BFS	$S \rightarrow B \rightarrow D \rightarrow G$	10

Conclusion: A* finds the optimal route because it considers both actual cost and heuristic estimate, whereas Greedy BFS considers only the heuristic.

7. Food Delivery Route Optimisation (Local Search)

A delivery partner starts at hub H, visits locations P, Q, R and S once each, and returns to H.

The distances (in km) are:

	H	P	Q	R	S
H	–	10	15	20	25
P	10	–	35	25	30
Q	15	35	–	30	20
	H	P	Q	R	S
R	20	25	30	–	15
S	25	30	20	15	–

- i. Find an initial tour using the Nearest Neighbour heuristic. ii. Improve the tour using Hill Climbing with 2-opt swaps. iii. Explain local maxima, plateaus and ridges, and how Simulated Annealing helps overcome them.

ANS:

i. Initial Tour Using Nearest Neighbour:

Start at H.

From H

Nearest location: $H \rightarrow P = 10$

Choose P.

From P

Remaining Q, R, S:

- $P \rightarrow Q = 35$
- $P \rightarrow R = 25$
- $P \rightarrow S = 30$

Choose R.

From R

Remaining Q, S:

- $R \rightarrow Q = 30$
- $R \rightarrow S = 15$

Choose S.

From S

Remaining Q: $S \rightarrow Q = 20$

Finally: $Q \rightarrow H = 15$

Therefore, initial tour: $H \rightarrow P \rightarrow R \rightarrow S \rightarrow Q \rightarrow H$

Cost:

$$\begin{aligned} &10 + 25 + 15 + 20 + 15 \\ &= \boxed{85 \text{ km}} \end{aligned}$$

ii. Hill Climbing with 2-opt:

The initial tour is: $H - P - R - S - Q - H$

Cost = **85 km**.

A 2-opt operation removes two edges and reconnects the route in another order.

For example, swapping/reversing sections can produce alternative tours such as:

$$H - P - Q - S - R - H$$

Cost: $10 + 35 + 20 + 15 + 20 = 100$

Another possible tour: $H - P - S - R - Q - H$

Cost: $10 + 30 + 15 + 30 + 15 = 100$

These are worse than 85 km.

The best 2-opt neighbourhood does **not improve** the 85 km solution.

Therefore Hill Climbing stops at:

$$\boxed{H \rightarrow P \rightarrow R \rightarrow S \rightarrow Q \rightarrow H}$$

with: $\boxed{85 \text{ km}}$

In fact, for the given distance matrix, **85 km is also the global optimum**.

iii. Local Maxima, Plateaus and Ridges:

Local Maximum :

A solution that is better than all its immediate neighbours but is **not the global best solution**.

Hill Climbing may stop here because it cannot find an immediately better move.

Plateau:

A region where several neighbouring states have the **same value**.

The algorithm may not know which direction to move.

Ridge:

A narrow path of improving solutions where a simple move cannot directly follow the best direction.

Hill Climbing can get stuck because its local moves do not allow it to move along the ridge effectively.

How Simulated Annealing Helps:

Simulated Annealing sometimes accepts a **worse solution temporarily**.

At high temperature, it accepts worse moves more frequently, allowing it to escape local maxima, plateaus and ridges.

As the temperature decreases, it becomes more selective and eventually behaves more like Hill Climbing.

Therefore:

Simulated Annealing helps escape local optima and find better global solutions.

14 .A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

ANS:

The customer wants to travel from one location to another using a cab-booking application. The customer can select different cab types such as Mini, Micro, Sedan, Shared, Prime, etc.

Type of Problem-Solving Agent:

A **Utility-Based Agent** is the most suitable.

Why?

The agent needs to consider several factors:

- Distance
- Travel time
- Cab type
- Fare
- Customer comfort
- Availability
- Traffic
- Customer preference

The best cab is therefore not necessarily the cheapest or fastest one. The agent can calculate a **utility value** for each available option and select the option with the highest utility.

Problem Formulation:

Initial State:

Customer is at the pickup location.

Goal State:

Customer reaches the destination.

Actions:

- Enter pickup and destination.
- Select cab type.
- Check available cabs.
- Compare fare/time/comfort.
- Book cab.

- Travel to destination.

Goal Test:

Customer reaches the selected destination successfully.

Path Cost:

The cost can include:

Fare + travel time + inconvenience + other factors.

Pseudocode:

BEGIN

Input pickup_location

Input destination

Input customer_preferences

Find available cab types

For each available cab:

 Calculate fare

 Estimate travel time

 Check comfort level

 Check availability

 Check traffic condition

Calculate utility value

 Utility =

 comfort_score

 - fare_factor

 - time_factor

 + preference_score

Select cab with highest utility

Display:

 Cab type

 Estimated fare

 Estimated travel time

 Driver/cab details

Ask customer for confirmation

IF customer confirms THEN

 Book selected cab

 Track cab location

 Navigate towards destination

IF destination reached THEN

Complete trip
 Calculate final fare
 Display trip summary
 END IF

ELSE
 Display other available cab options
 END IF

END

FINAL ANSWER: Utility-Based Agent

because it can evaluate multiple alternatives based on cost, time, comfort, availability and user preference, rather than simply choosing the first available cab.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem